Project Title: Historical Performance Analysis of Olympic Athletes

Project Objective

Analyze historical Olympic athlete data (1896–2016) to understand:

1. How medal performance differs between dominant and smaller countries.
2. Whether athlete physical attributes influence medal success in different sports.
3. Whether female participation in the Olympics has grown faster than male participation.

Initial Hypotheses

Hypothesis 1 — Country Performance

Countries with historically strong Olympic programs (USA, Russia/USSR, China) show **higher and increasing medal counts over time** compared to smaller nations.

Hypothesis 2 — Physical Attributes and Sport Success

Athlete physical characteristics (height, weight, age) influence medal outcomes depending on the sport.

Examples:

- **Height advantage:** Basketball, Volleyball, Swimming
- **Weight advantage:** Wrestling, Boxing
- **Age advantage:** Sprinting and Football favor younger athletes

Hypothesis 3 — Female Participation Growth

Female participation in the Olympics has **grown faster than male participation over time**.

Project Datasets

120 Years of Olympic History - **athelte_events.csv** with approximately **270k rows and 15 columns**

Region dataset – **noc_regions.csv** with **205 rows and three columns** mapping regions to NOCs

Data Preparation

The two datasets mentioned above were merged to map the team column in the Olympics Dataset with the country it is representing. To achieve this, a **LEFT JOIN was used in pandas**.

Data Cleaning

- Missing country mappings were identified after merging the athlete dataset with the NOC regions dataset. Some entries corresponded to boat names, unknown teams, or special Olympic delegations such as the Refugee Olympic Team. These were either corrected through manual mapping or removed when they did not correspond to real countries.
- Missing values in athlete attributes such as age, height, and weight were imputed using the median values within each sport to preserve sport-specific physical characteristics. Medal values were replaced with "No Medal" for athletes who did not win medals.
- All column names were made lowercase to ease analysis
- The duplicated athlete entries were identified and removed

Feature Engineering :

Following new features were added to the dataset to help in analysis:

- A Medal Flag feature was added to quantify the medals won by each athlete. It is a binary variable with a value of 1 for athlete winning gold, silver or bronze medals and 0 if they didn't win any medal. This helps in calculating Total Medals, Medal probability, and in building regression models.
- A decade column was added to the dataset to segment the performance of countries across the Olympic history into decades.
- A Body Mass Index (BMI) variable was calculated using athlete height and weight to capture overall body composition. This metric provides a standardized measure that allows comparison of athlete body types across different sports.

Exploratory Data Analysis:

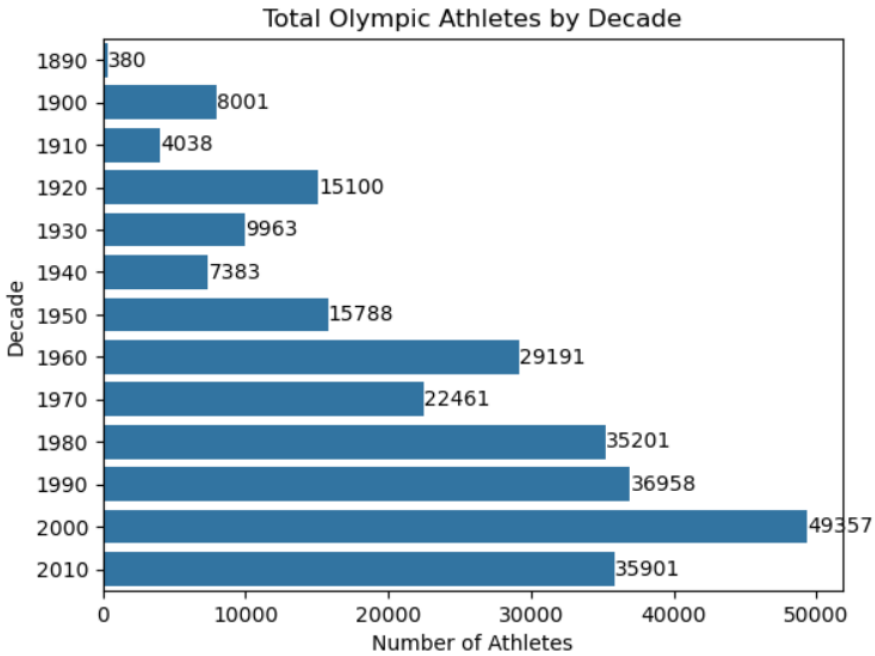
Total number of Athletes: **134,724**

Total Number of countries: **207**

Total Sports: **66**

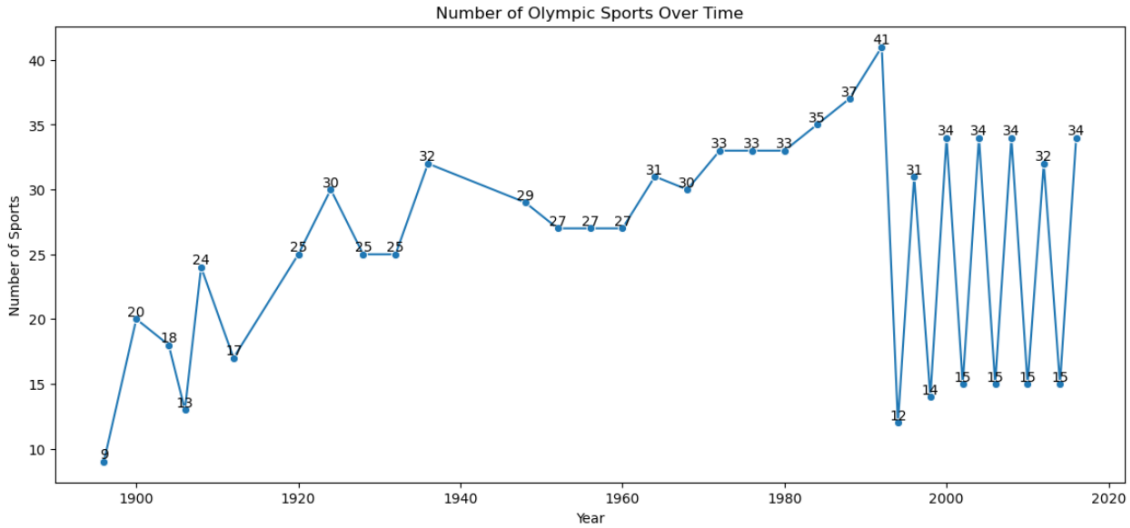
Total Olympic Games :**51**

Athlete Participation by Decade:



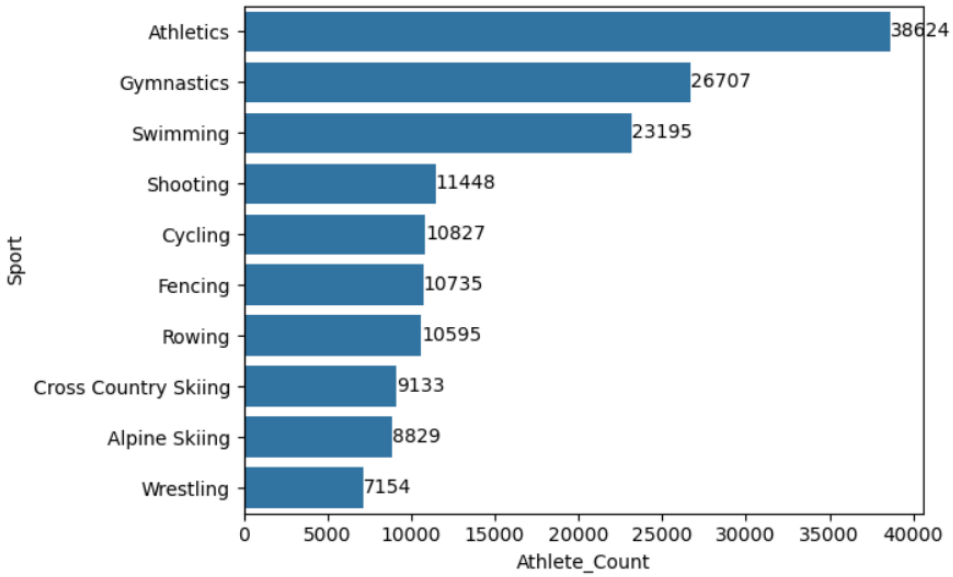
To explore the participation level of athletes, the above bar chart was plotted, which shows that the **participation levels have steadily increased** with the **2000 decade with the highest number of athletes.**

Number of Sports Over Olympic History:



The number of Olympic sports has steadily increased since the first modern Olympics in 1896. The expansion **accelerated after the mid-20th century** as new disciplines were introduced, **contributing to the growth in athlete participation and medal opportunities.**

TOP 10 most popular sports in Olympic History:



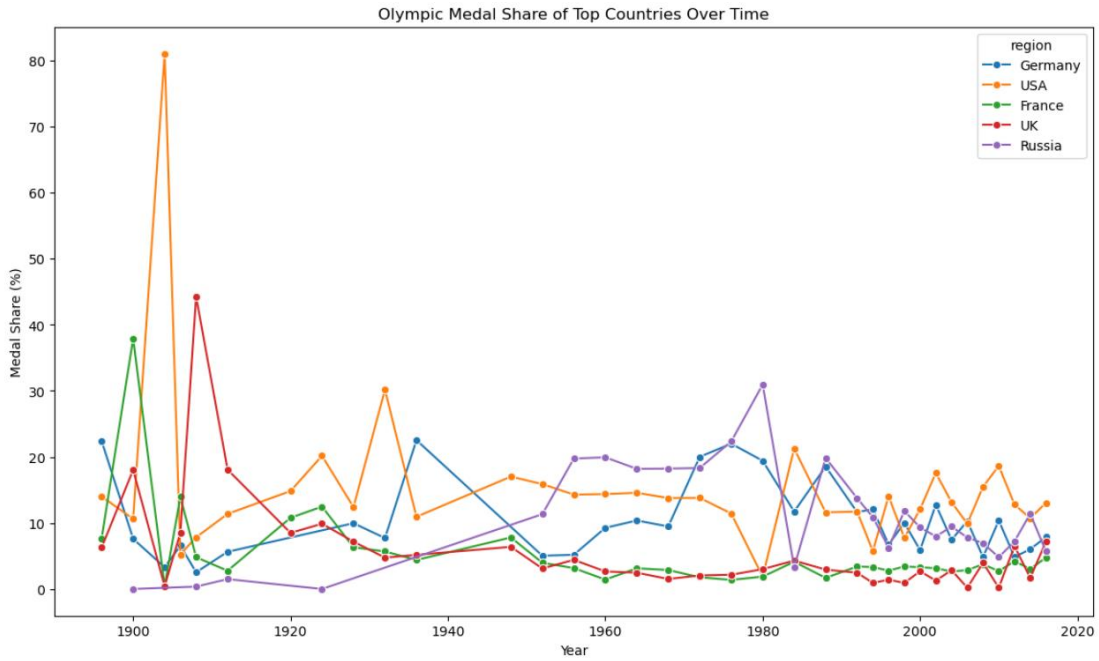
Athletics emerges as the most participated sport in Olympic history, with **over 38,000 athlete entries**, reflecting its inclusion across nearly all Olympic editions and the large number of track and field events

Analysis to test Hypothesis 1:

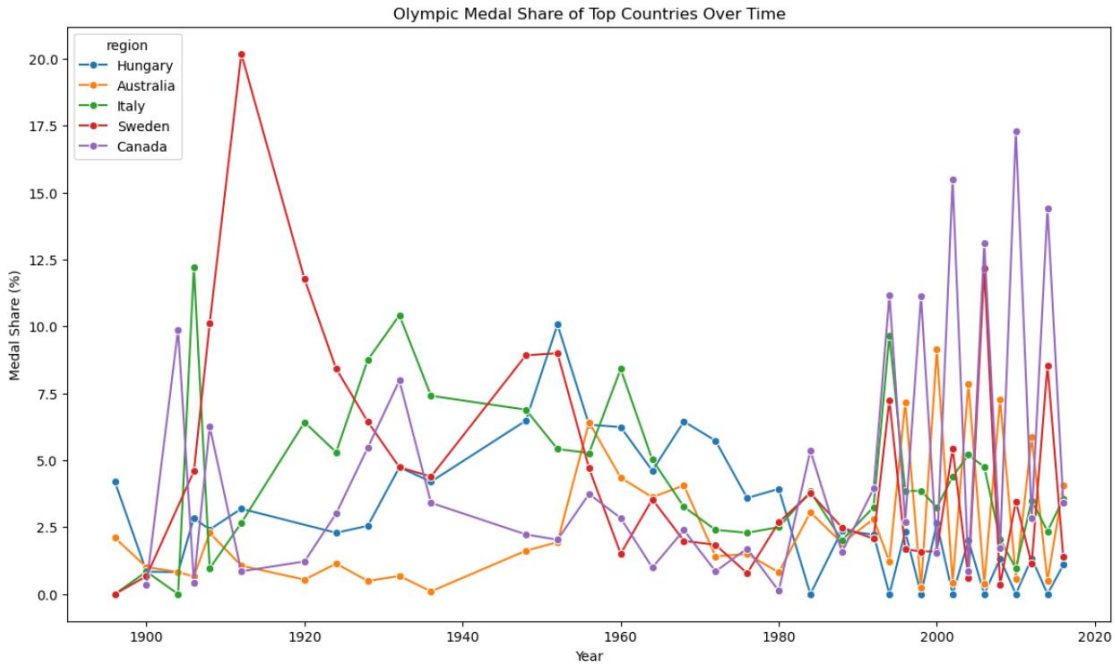
To test the hypothesis that countries with strong Olympic programs show increasing medal counts over time, the TOP 15 countries by total medal counts across Olympic history were filtered.

Then the medal share percentage of these Top 15 countries across Olympic history was plotted in groups of 5 for better clarity to investigate potential rise and fall in medal percentage.

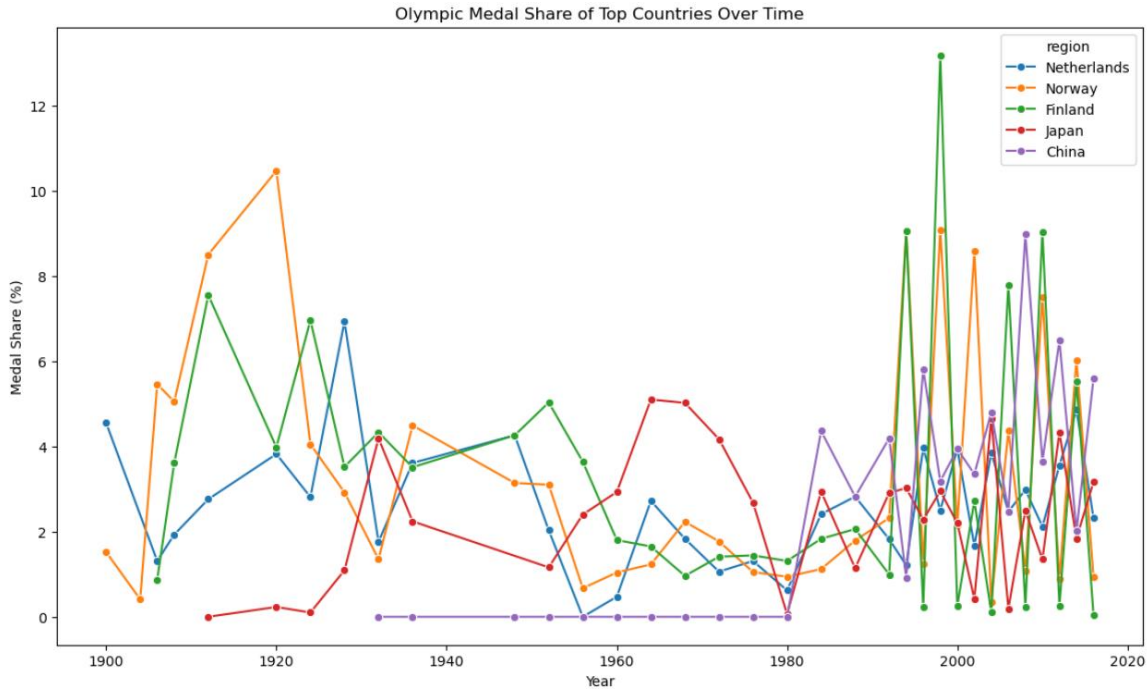
TOP 5 countries medal percentage across Olympic History:



Countries Ranked 6-10:



Countries Ranked 11- 15:



Based on the above analysis we get the below insights about medal trends:

1. ***The United States Shows Consistent Dominance:*** The United States demonstrates sustained dominance throughout Olympic history, consistently maintaining one of the highest medal shares despite increasing global competition.

2. ***Rise of Soviet/Russian Athletic Dominance During the Cold War:*** The Soviet Union emerged as a dominant Olympic competitor beginning in the mid-20th century, with medal shares rising significantly during the Cold War era.

3. ***China's Rapid Rise in the Modern Olympic Era:*** China shows one of the most significant modern increases in Olympic performance, with medal share rising rapidly from the 1980s onward.

4. ***Smaller Nations Show Irregular and Lower Medal Shares:*** Smaller nations display more volatile medal share trends, often achieving brief peaks but lacking the sustained dominance seen in larger Olympic programs.

5. ***Globalization of the Olympics Reduces Medal Concentration:*** Over time, medal shares become more evenly distributed as Olympic participation expands globally.

Conclusion on Hypothesis 1:

- The analysis **supports the hypothesis** that countries with historically strong Olympic programs—such as the United States, Soviet Union/Russia, and China—consistently **maintain higher medal shares over time**.
- In contrast, s**maller nations show more variable and generally lower medal shares**, reflecting differences in resources, sports infrastructure, and long-term investment in elite athletics.

Analysis to test Hypothesis 2:

To evaluate this hypothesis, sports were first categorized based on the **physical attributes generally associated with competitive advantage** in those sports. This step ensures that comparisons are made within contexts where a specific attribute is expected to play a meaningful role.

The sports were grouped into three categories:

Height-Advantage Sports

Sports where greater height typically provide a mechanical or reach advantage.

- Basketball
- Volleyball
- Rowing
- Swimming

Weight-Advantage Sports

Sports where greater body mass or strength contributes to competitive performance.

- Weightlifting
- Wrestling
- Judo
- Boxing

Age-Sensitive Sports

Sports where athletes often peak earlier due to agility, flexibility, or technical precision.

- Gymnastics
- Diving
- Figure Skating
- Swimming

For each category, athletes were divided into **medalists and non-medalists**, and the average physical attributes of the two groups were compared.

Results

Height Advantage Sports

The average height of medal-winning athletes in height-dominant sports was **184.28 cm**, while the average height of non-medalists was **181.45 cm**.

This represents a difference of approximately **3 cm**, suggesting that taller athletes may have a measurable advantage in sports where reach, leverage, or stroke length contribute to performance.

Weight Advantage Sports

The average weight of medalists in strength-based sports was **74.56 kg**, compared with **72.86 kg** for non-medalists.

This difference of approximately **1.7 kg** indicates that greater body mass may provide a competitive advantage in sports where strength and physical power are critical factors.

Age-Sensitive Sports

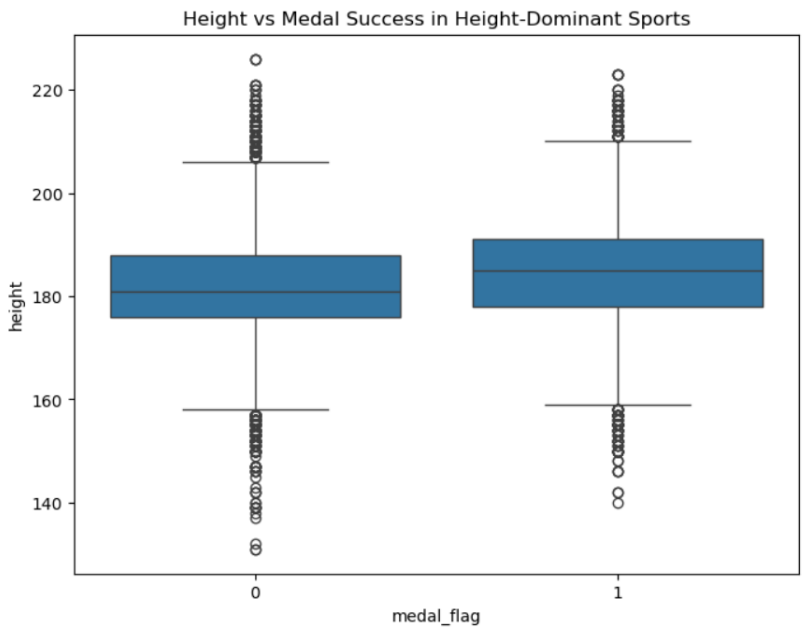
In sports typically associated with younger peak performance, the average age of medalists was **22.10 years**, compared with **21.86 years** for non-medalists.

The difference between the two groups was minimal, suggesting that age alone may not strongly differentiate medalists from non-medalists within these sports.

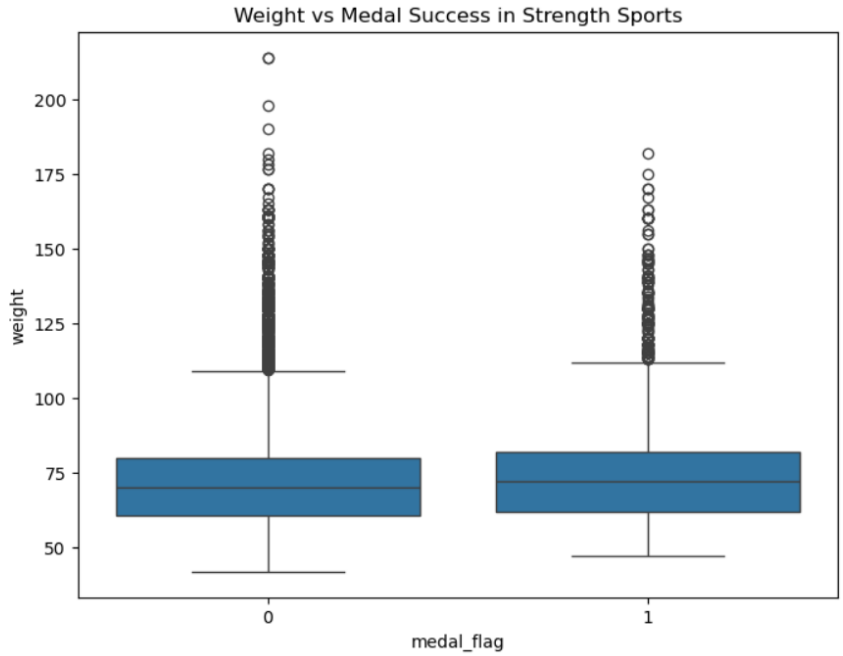
Visualization

To further examine these differences, **boxplots were generated comparing the distribution of height, weight, and age between medalists and non-medalists** within each sport category.

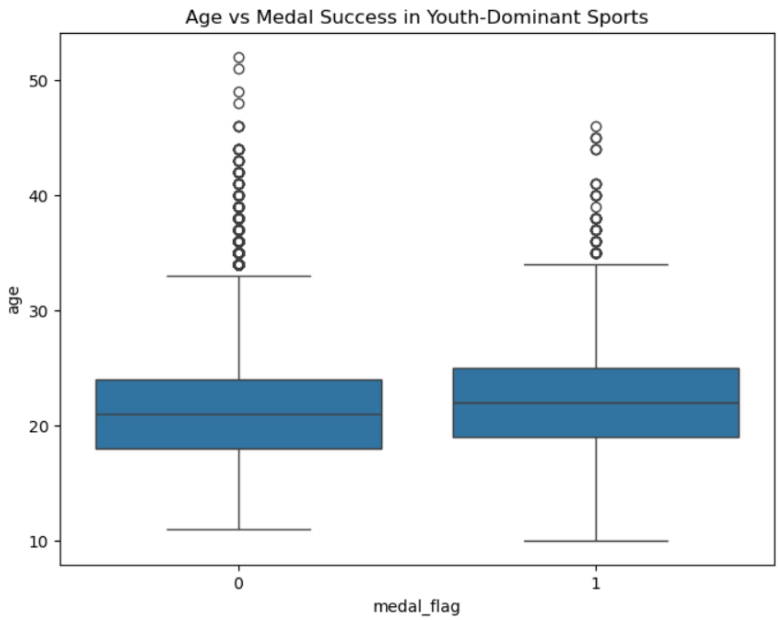
Boxplot of Height for Medalist and Non-medalist in Height advantage sports:



Boxplot of Weight for Medalist and Non-medalist in Weight advantage sports:



Boxplot of Age for Medalist and Non-medalist in Age advantage sports



These visualizations help illustrate:

- The spread of physical attributes across athletes
- Whether medalists consistently fall within higher or lower ranges of the attribute
- The presence of overlapping distributions between the two groups

The boxplots reinforce the observed patterns for **height and weight**, while the distributions for **age show considerable overlap between medalists and non-medalists**.

Statistical Validation

To assess whether the observed differences were statistically meaningful, **independent two-sample t-tests** were conducted comparing medalists and non-medalists for each physical attribute.

The results were as follows:

Attribute	p-value	Interpretation
Height	< 0.001	Statistically significant difference
Weight	< 0.001	Statistically significant difference
Age	< 0.001	Statistically significant but practically negligible

These results indicate that the differences in **height and weight** between medalists and non-medalists are statistically significant, meaning the observed differences are unlikely to be due to random variation.

The t-test for age differences between medalists and non-medalists produced **a statistically significant p-value due to the large number of observations in the dataset.** However, the actual difference in **average age between the two groups was very small** (approximately 0.24 years). This suggests that while the difference is statistically detectable, **age alone does not meaningfully influence medal outcomes in selected sports.**

Several insights emerge from this analysis:

1. **Height advantage is evident in certain sports**, particularly those involving reach, leverage, or stride length such as basketball, volleyball, and rowing.

2. **Weight appears to provide a competitive advantage in strength-oriented sports**, including wrestling, weightlifting, and boxing.

3. **Age does not appear to strongly differentiate medalists from non-medalists**, suggesting that while some sports favor younger athletes, age alone is not a reliable predictor of medal success.

4. **Physical advantages are sport-specific rather than universal**, highlighting the importance of analyzing athlete's characteristics within the context of each sport.

Conclusion on Hypothesis 2

The analysis provides **strong evidence supporting the role of physical attributes in Olympic success**, particularly height and weight in sports where these traits provide mechanical or strength advantages.

However, **age does not appear to significantly influence medal outcomes across selected sports**, suggesting that other factors such as training, experience, and technique may play a larger role.

Overall, the findings partially support the hypothesis:

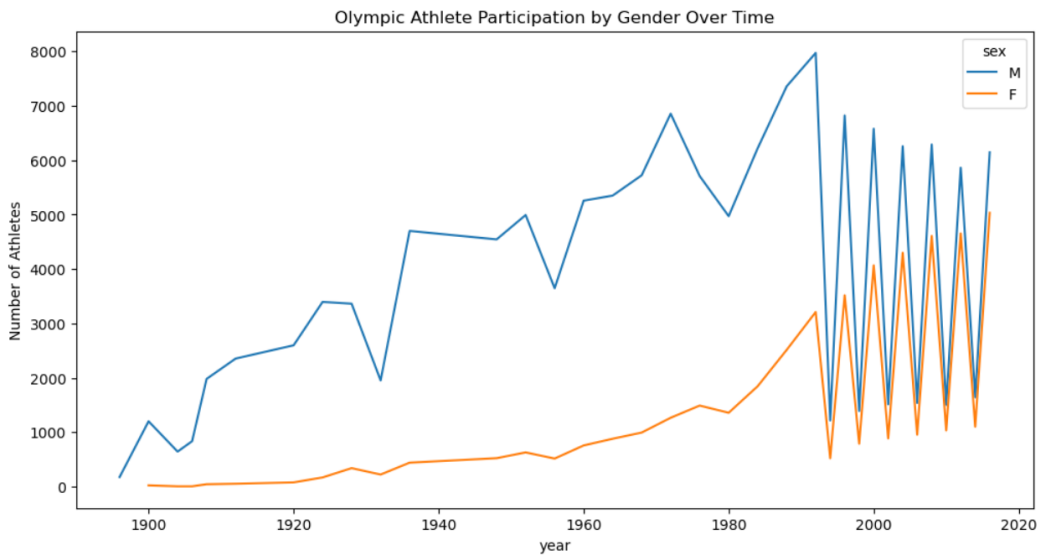
height and weight demonstrate measurable and statistically significant advantages in specific sports, while age shows a weaker relationship with medal success.

Analysis for Hypothesis 3:

To test this hypothesis, the analysis focused on examining **long-term trends in athlete participation by gender across Olympic years**. The objective was to determine whether female participation has increased at a faster rate compared to male participation over the history of the modern Olympic Games.

Visualize Participation Trends

A **line chart** was created to visualize how participation has changed for male and female athletes over time.



Insights from Number of Athletes Line plot:

1. Female participation was extremely low in early Olympic Games

The line plot shows that in the early years of the **Summer Olympics**, male athletes overwhelmingly dominated participation. Female participation remained extremely limited until the early 20th century.

- In the early Olympic Games (1896–1920), female athlete participation was **close to zero**.
- Male participation steadily increased during this period as the Olympic Games expanded globally.

This reflects the historical context where women had **very limited access to competitive sports** during the early years of the modern Olympics.

2. *Female participation began rising steadily after mid-20th century*

The line chart shows a **gradual increase in female athlete participation beginning around the 1950s–1960s**.

Key observations:

- Female participation grows steadily through the **1960–1980 period**.
- The number of female athletes increases **faster than the male athlete growth rate** during later decades.

This trend coincides with:

- Expansion of women's sporting events
- Increasing global participation in women's sports
- Greater emphasis on gender equality in international athletics.

Important note about the zig-zag pattern after 1990

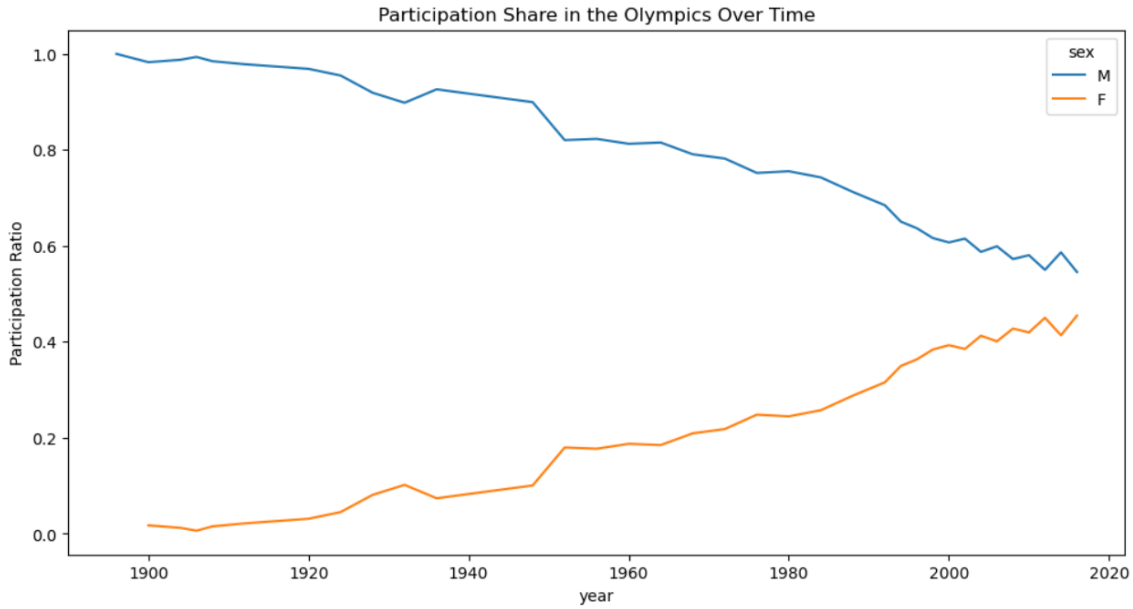
The alternating spikes visible in the first plot are caused by the **alternating scheduling of the Summer and Winter Olympic Games**, which were separated after the **1992 Olympic Games**.

This creates a pattern where athlete counts fluctuate between Winter and Summer Games rather than representing an actual drop in participation.

Absolute participation counts alone can be misleading because the **total number of Olympic athletes has also increased over time**.

To account for this, the **percentage share of female athletes per Olympic year** was calculated.

The below line plot is for percentage of male and female athletes in each Olympic.



Insights from Participation share Line Plot:

1. Female participation accelerated significantly after the 1990s

The participation share plot clearly shows that female representation **accelerates rapidly from the 1990s onward**.

Key pattern:

- Female participation share rises from **~25% in the 1980s to ~45% in recent Olympic Games**.
- Male participation share correspondingly declines from **~75–80% to nearly 55%**.

This indicates that female participation growth has **outpaced male participation growth**.

2. Olympic participation is approaching gender parity

The participation ratio plot highlights a major structural shift:

- Early Olympic Games were **almost entirely male dominated**
- Modern Olympic Games show **near gender balance**

Recent Olympics approach **almost equal representation of male and female athletes**.

This indicates that the Olympic movement has made significant progress toward **gender equality in sports participation**.

Conclusion on Hypothesis 3

The analysis strongly supports the hypothesis that **female participation in the Olympics has grown faster than male participation over time**.

Key conclusions:

- Female participation increased from **less than 5% in early Olympic Games to nearly 45% in recent Olympics**.
- The growth rate of female athletes has significantly outpaced male athlete growth over the past several decades.
- Modern Olympic Games are approaching **gender parity in athlete representation**.

Overall, the findings demonstrate a **long-term structural shift toward greater gender inclusion in the Olympic Games**, confirming that female participation has expanded at a significantly faster rate than male participation throughout Olympic history.

Final Summary of Findings

This analysis explored historical Olympic athlete data to examine trends in national performance, the influence of athlete physical attributes on success, and gender participation patterns across Olympic history.

For **Hypothesis 1**, the analysis of medal share across Olympic Games shows that countries with historically strong athletic infrastructures—such as the United States, Russia (including the former Soviet Union), and China—have consistently maintained a larger share of total Olympic medals compared to smaller nations. These countries demonstrate sustained dominance across decades, suggesting that factors such as investment in sports programs, training infrastructure, and athlete development systems contribute significantly to long-term Olympic success.

For **Hypothesis 2**, the analysis examined the relationship between athletes' physical attributes—height, weight, and age—and medal outcomes in sports where these traits are expected to provide competitive advantages. The results indicate that medalists tend to be **slightly taller in height-advantage sports** and **slightly heavier in strength-oriented sports**, with statistical testing confirming these differences as significant. However, age differences between medalists and non-medalists were minimal, suggesting that while physical characteristics like height and weight may influence success in certain sports, **age alone does not strongly determine medal outcomes** across the selected events.

For **Hypothesis 3**, the analysis of participation trends revealed a substantial increase in female athlete representation over time. Female participation has grown steadily from minimal representation in early Olympic Games to nearly equal participation in recent decades. The

participation share analysis confirms that **female athlete growth has outpaced male participation growth**, reflecting broader global efforts toward gender equality in sports.

Overall, the findings demonstrate three major long-term trends in Olympic history:

1. **Sustained dominance of historically strong sporting nations**,
2. **Sport-specific advantages associated with certain physical attributes**, and
3. **Rapid growth in female participation leading toward gender parity in the Olympics.**

Together, these results highlight how Olympic performance and participation patterns are shaped by **national investment in sports, athlete physical characteristics, and evolving global social and institutional changes in sport.**

Study Limitations

While the dataset provides extensive Olympic history, several limitations should be considered when interpreting the results.

Some athlete attributes such as **height and weight contain missing values**, which required imputation using sport-level median values. Although this approach preserves sport-specific characteristics, it may introduce **minor bias in the physical attribute analysis**.

Additionally, Olympic events include both **individual and team sports**, where medal outcomes may depend on factors beyond individual physical characteristics, such as team dynamics, strategy, and coaching quality.

Finally, the analysis focuses on a subset of sports when evaluating physical advantages. Results may therefore **not generalize to all Olympic sports**.

Future Analysis Scope

Future work could extend this analysis in several ways.

One potential direction is to build **predictive models** to estimate medal probability based on athlete attributes such as age, height, weight, sport category, and country representation.

Further analysis could also explore **country-level performance efficiency**, examining how effectively different nations convert athlete participation into medal outcomes.

These extensions would provide deeper insights into the factors influencing Olympic performance and athlete success.